CBSE Class X – Artificial Intelligence (417)

Previous Year Question Paper 2026 • Set 4 • Q.P. Code 104

Answer Key and Suggested Answers
Includes all objective questions, short answers, and long answers. For questions that ask candidates to answer only a specified number, answers to all listed questions are provided for revision.

Section A — Objective Type Questions

Q1. Employability Skills — answers to all six

1(i). A — Shut the front door. It gives a command/request, so it is imperative.

1(ii). C — Achievable. Dividing a large goal into smaller steps makes it more attainable.

1(iii). Time management is planning and organising how available time is used to complete tasks efficiently.

1(iv). D — Letters, numbers and special characters.

1(v). Entrepreneurs are born, not made. (This is the misconception stated in the question.)

1(vi). A — Population. “No poverty”, “Quality education” and “Reduced inequalities” are UN Sustainable Development Goals.

Q2. AI / Employability concepts — answers to all six

2(i). B — Classification. The model assigns a label such as apple, banana or orange.

2(ii). B — Context-dependent meaning. “Red” describes the face turning red due to anger, not simply the colour of an object.

2(iii). B — Ensure that choices made do not cause unintended harm.

2(iv). C — Statement 1 is correct, but Statement 2 is incorrect. Evaluation should use separate unseen data to test generalisation.

2(v). D — Reinforcement Learning. The system can learn suitable parking actions through rewards and penalties.

2(vi). C — Evaluation. This stage tests the model on data and assesses its performance.

Q3. AI concepts — answers to all six

3(i). B — False Negative. Failing to detect a real danger can lead to an accident.

3(ii). B — 0 to 255. An 8-bit byte represents 256 possible values, from 0 through 255.

3(iii). C — Association model. It finds relationships between items frequently purchased together.

3(iv). A — Intuition and values. Personal and social values can influence decisions made while designing AI.

3(v). C — Auto-generated captions on YouTube. Speech recognition converts spoken language into text.

3(vi). B — To evaluate the model’s accuracy on data not used for training.

Q4. AI concepts — answers to all six

4(i). C — False Positive (FP). The model predicts fire, but no fire is actually present.

4(ii). A — Both Assertion and Reason are true, and the Reason correctly explains the Assertion. Bioethics applies values and ethical principles to health, medicine and biological sciences.

4(iii). C — Predicting tomorrow’s temperature. Regression predicts a numerical value.

4(iv). D — The number of pixels in an image is known as its resolution (commonly expressed as width × height).

4(v). B — Correctly predicted positive observations divided by all predicted positive observations: Precision = TP / (TP + FP).

4(vi). B — Smart bot. Among the given options, a smart bot is the intended answer for a coded bot that can work with larger databases.

Q5. AI domains and evaluation — answers to all six

5(i). C — Statistical Data. Price comparison mainly involves comparing and analysing numerical prices and related product data.

5(ii). A — Both Assertion and Reason are true, and the Reason correctly explains the Assertion. Lowercasing makes “Hello” and “hello” consistent during text processing.

5(iii). D — Image Processing is a superset of Computer Vision (as intended by the options). Image processing modifies or enhances images; computer vision aims to interpret their content.

5(iv). True. Error measures the difference between a model’s predictions and the actual values.

5(v). B — Sentiment analysis. It identifies opinions or emotions expressed in customer reviews.

5(vi). C — 800. Correct predictions = TP + TN = 200 + 600 = 800.

Section B — Short Answers

Employability Skills (Q6–Q10) — suggested answers in approximately 20–30 words

Q6. Write the 7 C's of communication.

Clear, Concise, Concrete, Correct, Coherent, Complete and Courteous. These principles help make communication understandable, accurate, logically organised and respectful.

Q7(a). What is emotional intelligence?

Emotional intelligence is the ability to recognise, understand and manage one's own emotions and to recognise and respond appropriately to other people's emotions.

Q7(b). Name any two skills included in emotional intelligence.

Two skills are self-awareness and self-regulation. Other acceptable examples include empathy, motivation and social skills.

Q8. How do ICT skills help in day-to-day activities?

ICT skills help us communicate, search for information, prepare documents, attend online classes, make digital payments and store or share information efficiently.

Q9. State qualities of a successful entrepreneur.

A successful entrepreneur is creative, confident, hardworking, persistent, willing to take calculated risks, adaptable and able to solve problems and make decisions.

Q10. Define Sustainable Development.

Sustainable development meets the needs of the present without compromising the ability of future generations to meet their own needs.

AI subject questions (Q11–Q16) — suggested answers in approximately 20–30 words

Q11. Sector-based vs value-based ethical frameworks, with examples.

Sector-based frameworks address ethics in a particular field, such as healthcare AI. Value-based frameworks apply broad values, such as fairness, privacy or avoiding harm, across applications.

Q12. Give two characteristics of a Classification Model.

A classification model predicts discrete categories or labels, such as spam/not spam. It learns patterns from labelled examples and assigns new inputs to a class.

Q13. How do computers store RGB images?

Each pixel stores three channel values—Red, Green and Blue. In an 8-bit-per-channel image, each channel ranges from 0 to 255, giving about 16.7 million combinations.

Q14. Two differences between supervised and unsupervised learning.

Supervised learning uses labelled data and learns to predict known targets. Unsupervised learning uses unlabelled data to discover patterns, clusters or relationships.

Q15. Explain train-test split.

The dataset is divided into training and testing sets. The model learns from the training set, then its performance is measured on the separate test set.

Q16. Stemming vs lemmatization; process "Wolves".

Stemming mechanically removes word endings and may produce a non-dictionary stem. Lemmatization uses vocabulary and grammar; "wolves" is lemmatized to "wolf". A stemmer may return "wolv" depending on its algorithm.

Section B — Long Answers

Answer any three in the examination; all five are supplied here for revision.

Q17. Differentiate between AI, ML and Deep Learning. Show their relationship.

Artificial Intelligence (AI) is the broad field of creating systems that perform tasks requiring human-like intelligence, such as reasoning, language understanding or perception. Machine Learning (ML) is a subset of AI in which systems learn patterns from data rather than relying only on fixed rules. Deep Learning (DL) is a subset of ML that uses neural networks with multiple layers to learn complex patterns, often from large datasets. Thus, DL is inside ML, and ML is inside AI.

Q18. Identify the most appropriate AI domain for each scenario and justify.

(A) Natural Language Processing (NLP): the platform must translate essays and analyse grammar, content and writing style. NLP processes and interprets human language, enabling translation, text analysis and feedback. (B) Computer Vision: the system scans vehicles and classifies them as two-wheelers or four-wheelers from visual input. Computer Vision analyses images/video and recognises objects or categories.

Q19. Confusion matrix, true negatives and precision.

Treat “cyber attack” as the positive class. The model correctly identifies 1,000 attacks (TP), correctly identifies 250 non-attacks (TN), falsely flags 200 non-attacks as attacks (FP), and misses 50 actual attacks (FN).

Actual \ Predicted	Attack	No attack
Attack	TP = 1000	FN = 50
No attack	FP = 200	TN = 250

(B) True Negatives = 250.
(C) Precision = TP / (TP + FP) = 1000 / (1000 + 200) = 1000 / 1200 = 0.8333 = **83.33%**.

Q20(A). Expand and define CNN and ANN.

CNN stands for Convolutional Neural Network. It is a neural network especially useful for image data; convolutional filters learn visual features such as edges, shapes and objects. ANN stands for Artificial Neural Network. It is a computing model inspired by interconnected neurons, organised into layers that process inputs and learn patterns by adjusting connection weights.

Q20(B). Identify Box 1 and Box 2 in the neural-network diagram.

Box 1: Input layer (the first set of neurons receiving input data). Box 2: Hidden layer (intermediate neurons that process the information). Box 3, shown for reference, is the output layer.

Q21. Apply all four Bag of Words steps to create a document vector table.

Step 1 — Text preparation: lowercase the documents and remove punctuation: D1 = “data science requires information”; D2 = “information analysis requires data”. Step 2 — Tokenisation: split each document into words. Step 3 — Build vocabulary of unique words in order: data, science, requires, information, analysis. Step 4 — Count each vocabulary word in each document to create the document vectors.

Document	data	science	requires	information	analysis
D1	1	1	1	1	0
D2	1	0	1	1	1

Note: This table keeps all distinct words, including “requires”. If a classroom-specific stop-word list is required, the vocabulary and vectors may differ.

Important note on interpretation

The answers are suggested model answers based on the attached question paper. For Q5(i), the paper’s options are unusual because a price-comparison website may also use data analytics; “Statistical Data” is the closest intended

option among those listed. In Q5(iii), the answer follows the intended distinction in the provided options. Awarding may depend on the official marking scheme.